FOOD SAFETY AND STANDARDS (CONTAMINANTS, TOXINS AND RESIDUES) REGULATIONS, 2011

CHAPTER 1
GENERAL

1.1 : Short title and commencement-

1.1.1 : These regulations may be called the Food Safety and Standards (Contaminants, toxins and Residues) Regulations, 2011.

1.1.2 : These regulations shall come into force on or after 5th August, 2011.

1.2 : Definitions-

1.2.1 : In these regulations unless the context otherwise requires:

1. "Crop contaminant" means any substance not intentionally added to food, but which gets added to articles of food in the process of their production (including operations carried out in crop husbandry, animal husbandry and veterinary medicine), manufacture, processing, preparation, treatment, packing, packaging transport orholding of articles of such food as a result of environmental contamination

CHAPTER 2
CONTAMINANTS, TOXINS AND RESIDUES

2.1 : METAL CONTAMINANTS

[15][2.1.1

1. Chemicals described in monographs of the Indian Pharmacopoeia when used in foods, shall not contain metal contaminants beyond the limits specified in the appropriate monographs of the Indian Pharmacopoeia for the time being in force.
2. Notwithstanding anything contained in clause (1) above, no article of food specified in column (2) of the table below shall contain any metal specified in excess of the quantity specified in column (3) of the said table:

Name of metal contaminant	Article of food	Parts per Million (mg/kg or mg/L)
(1)	(2)	(3)
1. Lead	Agar	5.0
	Alginic acid	5.0

	All types of sugars, sugar syrup, invert sugar and direct consumption coloured sugars with sulphated ash content exceeding 1.0 percent	5.0
	Alumina used in preparation of lake colour	10
	Aluminium lake of Sunset Yellow FCF	10
	Ammonium hydrogen carbonate	2.0
	Anhydrous dextrose and dextrose monohydrate, refined white sugar (sulphated ash content not exceeding 0.03 per cent)	0.5
	Annatto	10
	Ascorbic acid	2.0
	Ascorbyl palmitate	2.0
	Aspertame (Aspartyl phenyl alanine methyl ester)	10
	Assorted subtropical fruits, edible peel	0.1
	Assorted subtropical fruits, inedible peel	0.1
	Baking powder	10
	Benzoic acid	2.0
	Berries and other small fruits	0.2
	Beta-apo-8'-carotenal	2.0
	Beta-carotene	10
	Bivalve molluscs	1.5
	Brassica vegetables excluding Kale	0.3
	Brewed vinegar and synthetic vinegar	0.01
	Brilliant blue FCF	10
	Bulb vegetables	0.1
	Butylated hydroxyanisole	2.0
	Calcium alginate	5.0
	Calcium propionate	5.0
	Canned carrots	1.0
	Canned green beans and canned wax beans	1.0
	Canned green peas	1.0
	Canned mushrooms	1.0
	Canned palmito	1.0
	Canned sweetcorn	1.0
	Canned tomatoes	1.0
	Canned asparagus	1.0
	Canned chestnuts and canned chestnut purée	1.0

	Canned fish, canned meats, edible gelatin,meat extracts and hydrolysed protein, dried or dehydrated vegetables (other than onions)	5.0
	Canned fruit cocktail	1.0
	Canned grapefruit	1.0
	Canned mandarin oranges	1.0
	Canned mangoes	1.0
	Canned mature processed peas	1.0
	Canned pineapple	1.0
	Canned raspberries	1.0
	Canned strawberries	1.0
	Canned tropical fruit salad	1.0
	Caramel	5.0
	Carbonated water , expressed in mg/L	10
	Carmoisine	10
	Carrageenan	5.0
	Cattle, edible offal of	0.5
	Cephalopods	1.0
	Cereal grains, except buckwheat, canihua and quinoa	0.2
	Chlorophyll	10
	Citric acid	0.5
	Citrus fruits	0.1
	Cocoa powder	5.0 on dry fat free substance basis
	Concentrated soft drinks (but not including concentrates used in the manufacture of soft drinks)	0.5
	Concentrates used in the manufacture of soft drinks, lime juice and lemon juice	2.0
	Corned beef, Luncheon meat, Cooked ham, Chopped meat, Canned chicken, Canned mutton and Goat meat and other related meat products	2.5
	Crustaceans	0.5
	Dehydrated onions, dried herbs and spices, curry powder and mix masalas, flavourings, alginic acid, alignates, agar, carrageen and similar products derived from seaweed	10 on dry matter basis
	Dicalcium phosphate	4.0
	Dodecyl gallate	2.0
	Edible fats and oils (edible fats and oils not covered by individual standards)	0.1

	Edible molasses, caramel liquid, solid glucose and starch conversion products with a sulphated ash content exceeding 1.0 per cent	5.0
	Edible oils and fats	0.5
	Erythrosine	10
	Ethylester of Beta-apo-8'-carotenoic acid	2.0
	Fast green FCF	10
	Fish	0.3
	Food colours other than caramel	10 on dry colouring matter basis
	Foods not specified	2.5
	Fruit and vegetable juice (including tomato juice, but not including lime juice and lemon juice)	1.0
	Fruit Juices (including nectars; ready to drink)	0.05
	Fruiting vegetables other than cucurbits(excluding mushrooms)	0.1
	Fruiting vegetables, cucurbits	0.1
	Fumaric acid	2.0
	Gaur gum	2.0
	Glycerol esters of Wood rosin	1.0
	Gum Arabic or Acacia gum	3.0
	Gum ghatti	5.0
	Gum karaya	2.0
	Hard boiled sugar confectionery	2.0
	Ice-cream, iced lollies and similar frozen confections	1.0
	Indigo carmine	10
	Infant formula (ready to use)	0.02
	Infant milk substitutes and Infant foods	0.2
	Iron fortified common salt	2.0
	Jam (fruit preserves) and jellies	1.0
	L (+) -Tartaric acid	2.0
	Lactic acid	2.0
	Leafy vegetables (including brassica leafy vegetables but excluding spinach)	0.3
	Legume vegetables	0.2
	Liquid pectin, chemicals not otherwise specified, used as ingredients or in the preparation or processing of food	10
	Malic acid	2.0
	Mango chutney	1.0

	Margarine	0.1
	Meat of cattle, sheep and pig (also applies to fat from meat)	0.1
	Milks (Concentration factor shall be applied to partially or wholly dehydrated milks)	0.02
	Minarine **(Low Fat Spread)**	0.1
	Mineral Oil (High viscosity)	1.0
	Mineral Oil (Low viscosity)	1.0
	Monosodium L-glutamate	1.0
	Named Animal fats (lard, rendered pork fat, premier jus **(suet)** and edible tallow)	0.1
	Natural mineral water, expressed in mg/L	0.01
	Octyl gallate	2.0
	Olive oil, Virgin olive oil, Extra virgin olive oil, Ordinary virgin olive oil, Refined olive oil, Refined olive pomace oil and Olive pomace oil	0.1
	Other vegetables	2.5
	Packaged drinking water (other than mineral water), expressed in mg/L	0.01
	Pectin	2.0
	Phosphoric acid	4.0
	Pickled cucumbers (Cucumber pickles)	1.0
	Pig, edible offal of	0.5
	Polyglycerol esters of fatty acids	2.0
	Polyglycerol esters of interesterified ricinoleic acid	2.0
	Pome fruits	0.1
	Ponceau 4R	10
	Potassium iodate	10
	Potassium metabisulphite	2.0
	Potassium nitrate	2.0
	Potassium nitrite	2.0
	Poultry fats	0.1
	Poultry meat	0.1
	Poultry, edible offal of	0.5
	Processed tomato concentrates	1.5
	Propyl gallate	2.0
	Propylene glycol	2.0
	Pulses	0.2
	Raw sugars except those sold for direct consumption or used for manufacturing purpose other than the manufacture of refined sugar	5.0
	Riboflavin	20
	Root and tuber vegetables	0.1

	Saccharin sodium	10
	Salt, food grade	2.0
	Secondary milk products (as consumed)	0.02
	Sodium alginate	5.0
	Sodium ascorbate	2.0
	Sodium benzoate	2.0
	Sodium carboxymethyl cellulose	2.0
	Sodium carboxymethyl cellulose, enzyme hydrolysed	3.0
	Sodium hydroxide	2.0
	Sodium metabisulphite	2.0
	Sodium propionate	5.0
	Solid pectin	50
	Sorbic acid	2.0
	Sorbitol	1.0
	Steviol glycoside	1.0
	Stone fruits	0.1
	Sucralose	10
	Sulphur dioxide	5.0
	Sunset yellow	10
	Sunset yellow dye used in preparation of lake colour	10
	Synthetic food colour-preparation and mixtures	10
	Table olives	1.0
	Tartrazine	10
	Tea	5.0 on dry matter basis
	Titanium dioxide	2.0
	Tragacanth gum	2.0
	Trisodium citrate	2.0
	Turmeric whole and powder	10
	Vegetable Oils, crude (oils of arachis **(Groundnut),** babasu, coconut, cotton seed, grape seed, maize, mustard seed, palm kernel, palm, rape seed, safflower seed, sesame seed, soya bean, and sunflower seed, and palm olein, stearin and superolein and other oils but excluding cocoa butter)	0.1
	Vegetable Oils, edible (oils of arachis **(Groundnut**), babasu, coconut, cotton seed, grape seed, maize, mustard seed, palm kernel, palm, rape seed, safflower seed, sesame seed, soya bean, and sunflower seed, and palm olein, stearin and superolein and other oils but	0.1

	excluding cocoa butter)	
	Wine	0.2
	Yeast and yeast products	5.0 on dry matter basis
2. Copper	Ammonium hydrogen carbonate	5.0
	Annatto	30
	Brewed vinegar and synthetic vinegar	0.01
	Caramel	20
	Carbonated water , expressed in mg/L	1.5
	Chicory-dried or roasted, coffee beans, flavourings/pectin liquid	30
	Chlorophyll	30
	Cocoa powder	70 on fat free substance basis
	Colouring matter	30 on dry colouring matter basis
	Concentrates for soft drinks	20
	Edible gelatin	30
	Foods not specified	30
	Hard boiled sugar confectionery	5.0
	Infant milk substitute and Infant foods	15 (But not less than 2.8)
	Iron fortified common salt	2.0
	Juice of orange, grape, apple, tomato, pineapple and lemon	5.0
	Mineral water , expressed in mg/L	1.0
	Olive oil, Virgin olive oil ,Extra virgin olive oil, Ordinary virgin olive oil, Refined olive oil, Refined olive pomace oil and Olive pomace oil	0.1
	Packaged drinking water (other than mineral water), expressed in mg/L	0.05
	Solid Pectin	300
	Polyglycerol esters of fatty acids	25
	Polyglycerol esters of Interesterified ricinoleic acid	25
	Pulp and pulp products of any fruit	5.0
	Soft drinks excluding concentrates and Carbonated Water , expressed in mg/L	7.0
	Tea	150
	Toddy	5.0

	Tomato ketchup	50 on dried total solids basis
	Tomato puree, paste, powder, and cocktails	100 on dried tomato solids
	Turmeric whole and powder	5.0
	Vegetables	30
	Yeast and yeast products	60 on dry matter basis
3. Arsenic	Agar	3.0
	Alginic acid	3.0
	Alumina used in preparation of lake colour	1.0
	Aluminium lake of Sunset Yellow FCF	1.0
	Ammonium hydrogen carbonate	0.6
	Annatto	3.0
	Ascorbyl palmitate	3.0
	Aspertame (Aspartyl phenyl alanine methyl ester)	3.0
	Benzoic acid	3.0
	Beta –apo-8'-carotenal	3.0
	Beta-carotene	3.0
	Brewed vinegar and synthetic vinegar	0.1
	Brilliant blue FCF	3.0
	Butylated hydroxyanisole	3.0
	Calcium alginate	3.0
	Caramel	3.0
	Carbonated water, expressed in mg/L	0.25
	Carmoisine	3.0
	Carrageenan	3.0
	Chicory-dried or roasted	4.0
	Chlorophyll	3.0
	Citric acid	3.0
	Dehydrated onions, edible gelatin, liquid pectin	2.0
	Dicalcium phosphate	3.0
	Dodecyl gallate	3.0
	Dried herbs, finings and clearing agents, solid pectin all grades, spices	5.0
	Edible fats and oils (edible fats and oils not covered by individual standards)	0.1
	Erythrosine	3.0
	Ethylester of Beta-apo-8'-carotenoic acid	3.0
	Fast Green FCF	3.0
	Fish and Crustaceans	76

	Food colouring other than synthetic colouring	5.0 on dry colouring matter basis
	Foods not specified	1.1
	Fumaric acid	3.0
	Gaur gum	3.0
	Glycerol esters of wood rosin	3.0
	Gum Arabic or Acacia gum	2.0
	Gum Ghatti	3.0
	Gum Karaya	3.0
	Hard boiled sugar confectionery	1.0
	Ice-cream, iced lollies and similar frozen confections	0.5
	Indigo carmine	3.0
	Infant milk substitute and Infant foods	0.05
	Iron fortified common salt	1.0
	Juice of orange, grape, apple, tomato, pineapple and lemon	0.2
	L (+)- Tartaric acid	3.0
	Malic acid	3.0
	Margarine	0.1
	Milk	0.1
	Minarine **(Low Fat Spread)**	0.1
	Mineral Oil (High viscosity)	1.0
	Mineral Oil (Low viscosity)	1.0
	Molluscs	86
	Monosodium L-glutamate	2.0
	Named Animal fats (lard, rendered pork fat, premier jus**(suet)** and edible tallow)	0.1
	Natural mineral water, expressed in mg/L	0.01
	Octyl gallate	3.0
	Olive oil, Virgin olive oil ,Extra virgin olive oil, Ordinary virgin olive oil, Refined olive oil, Refined olive pomace oil and Olive pomace oil	0.1
	Packaged drinking water (other than mineral water) , expressed in mg/L	0.01
	Pectin	5.0
	Phosphoric acid	2.0
	Polyglycerol esters of fatty acids	3.0
	Polyglycerol esters of interesterified ricinoleic acid	3.0
	Ponceau 4R	3.0
	Potassium iodate	3.0
	Potassium nitrate	3.0
	Potassium nitrite	3.0

	Preservatives, anti-oxidants, emulsifying and stabilising agents and synthetic food colours	3.0 on dry matter basis
	Propyl gallate	3.0
	Propylene glycol	3.0
	Pulp and pulp products of any fruit	0.2
	Riboflavin	5.0
	Saccharin sodium	2.0
	Sodium alginate	3.0
	Sodium ascorbate	3.0
	Sodium benzoate	3.0
	Sodium carboxymethyl cellulose	3.0
	Sodium propionate	3.0
	Soft drink intended for consumption after dilution except carbonated water	0.5
	Sorbic acid	3.0
	Sorbitol	3.0
	Steviol glycoside	1.0
	Sucralose	3.0
	Sulphur dioxide	3.0
	Sunset yellow	3.0
	Sunset yellow dye used in preparation of lake colour	3.0
	Synthetic food colour-preparation and mixtures	3.0
	Tartrazine	3.0
	Titanium dioxide	1.0
	Tragacanth gum	3.0
	Trisodium citrate	3.0
	Turmeric whole and powder	0.1
	Vegetables	1.1
	Vegetable oils, crude (oils of arachis **(Groundnut),** babasu, coconut, cotton seed, grape seed, maize, mustard seed, palm kernel, palm, rapeseed, safflower seed, sesame seed, soya bean, and sunflower seed, and palm olein, stearin and superolein).	0.1
	Vegetable oils, edible (oils of arachis **(Groundnut)**, babasu, coconut, cotton seed, grape seed, maize, mustard seed, palm kernel, palm, rapeseed, safflower seed, sesame seed, soya bean, and sunflower seed, and palm olein, stearin and superolein).	0.1
4.Tin	Canned (citrus fruits, stone fruits, vegetables, fruit cocktail, mangoes, pineapple, raspberries, strawberries, tropical fruit salad).	250
	Canned beverages	150

	Canned chestnuts and chestnut puree	250
	Canned fish products	200
	Canned foods other than beverages	250
	Canned mushrooms	250
	Canned tomatoes	250
	Cooked cured chopped meat (for products in other containers)	50
	Cooked cured chopped meat (for products in tinplate containers)	250
	Cooked cured ham (for products in other containers)	50
	Cooked cured ham (for products in tinplate containers	200
	Cooked cured pork shoulder (for products in other containers)	50
	Cooked cured pork shoulder (for products in tinplate containers)	200
	Corned beef (for products in other containers)	50
	Corned beef (for products in tinplate containers)	200
	Corned beef, Luncheon meat, Cooked ham, Chopped meat, Canned chicken, Canned mutton and Goat meat	250
	Foods not specified	250
	Hard boiled sugar confectionery	5.0
	Infant milk substitute and Infant foods	5.0
	Jam, Jellies and Marmalade	250
	Juice of orange, apple, tomato, pineapple and lemon	250
	Luncheon meat (for products in other containers)	50
	Luncheon meat (for products in tinplate containers)	200
	Mango Chutney	250
	Pickled cucumber	250
	Processed and canned food products	250
	Processed tomato concentrates	250
	Pulp and pulp products of any fruit	250
	Table Olives	250
	Turmeric whole and powder	0.01
5. Cadmium	Bivalve Molluscs	2.0
	Brassica vegetables	0.05
	Bulb vegetables	0.05
	Carrageenan	1.5
	Cephalopods	2.0
	Cereal grains, except buckwheat, canihua and	0.1

	Quinoa (excluding wheat and rice; and bran and germ)	
	Crustaceans	0.5
	Fish	0.3
	Foods not specified	1.5
	Fruiting vegetables other than cucurbits (excluding tomatoes and edible fungi)	0.05
	Fruiting vegetables, cucurbits	0.05
	Infant milk substitute and Infant foods	0.1
	Leafy vegetables	0.2
	Legume vegetables	0.1
	Natural mineral water, expressed in mg/L	0.003
	Other vegetables	1.5
	Packaged drinking water (other than mineral water), expressed in mg/L	0.003
	Potato, peeled	0.1
	Pulses, excluding soybean dry	0.1
	Rice, polished	0.4
	Root and tuber vegetables, excluding potato and celeriac	0.1
	Salt, food grade	0.5
	Stalk and stem vegetables	0.1
	Turmeric whole and powder	0.1
	Wheat	0.2
6. Mercury	Alumina used in preparation of lake colour	1.0
	Aluminium lake of Sunset yellow FCF	1.0
	Caramel	0.1
	Carrageenan	1.0
	Fast green FCF	0.01
	Fish	0.5
	Foods not specified	1.0
	Natural mineral water, expressed in mg/L	0.001
	Non-predatory fish, crustaceans, cephalopods, molluscs	0.5
	Packaged drinking water (other than mineral water), expressed in mg/L	0.001
	Predatory fish (Tuna, Marlin, Sword Fish, Elasmobranch)	1.0
	Salt, food grade	0.1
	Sodium hydroxide	1.5
	Titanium oxide	1.0
	Vegetables	1.0
7. Methyl Mercury (Calculated as the element)	All foods	0.25

8. Chromium	All fishery products	12
	Brilliant blue FCF	50
	Fast green FCF	50
	Gelatin	10
	Mineral water, expressed in mg/L	0.05
	Packaged drinking water (other than mineral water), expressed in mg/L	0.05
	Refined sugar	0.02
	Vegetables	1.0
9. Nickel	All hydrogenated, partially hydrogenated, interesterified vegetable oils and fats such as vanaspati, table margarine, bakery and industrial margarine, bakery shortening, fat spread and partially hydrogenated margarine, bakery shortening, fat spread and partially hydrogenated soyabean oil	1.5
	Mineral water, expressed in mg/L	0.02
	Packaged drinking water (other than mineral water), expressed in mg/L	0.02
	Sorbitol	2.0
	Vegetables	1.0
10.Selenium	Mineral water , expressed in mg/L	0.05
	Packaged drinking water (other than mineral water), expressed in mg/L	0.01
	Potassium metabisulphite	5.0
	Sodium metabisulphite	5.0
	Sulphur dioxide	20
11.Antimony	Mineral water , expressed in mg/L	0.005
	Packaged drinking water (other than mineral water), expressed in mg/L	0.005
	Titanium dioxide	2.0
	Vegetables	1.0”]

Table